

Algorithm 3

Gaussian Distributions

Speech and silence segmentation

Frame 25 ms. Hop 10 ms.

Minimum silence duration: 200 ms

Name / Student ID: pending

Algorithm 3 workflow

1. Read WAV with 25/10 ms framing



2. Compute and normalize STE



3. Estimate training mean and standard deviation



4. Create candidate speech using the Gaussian threshold

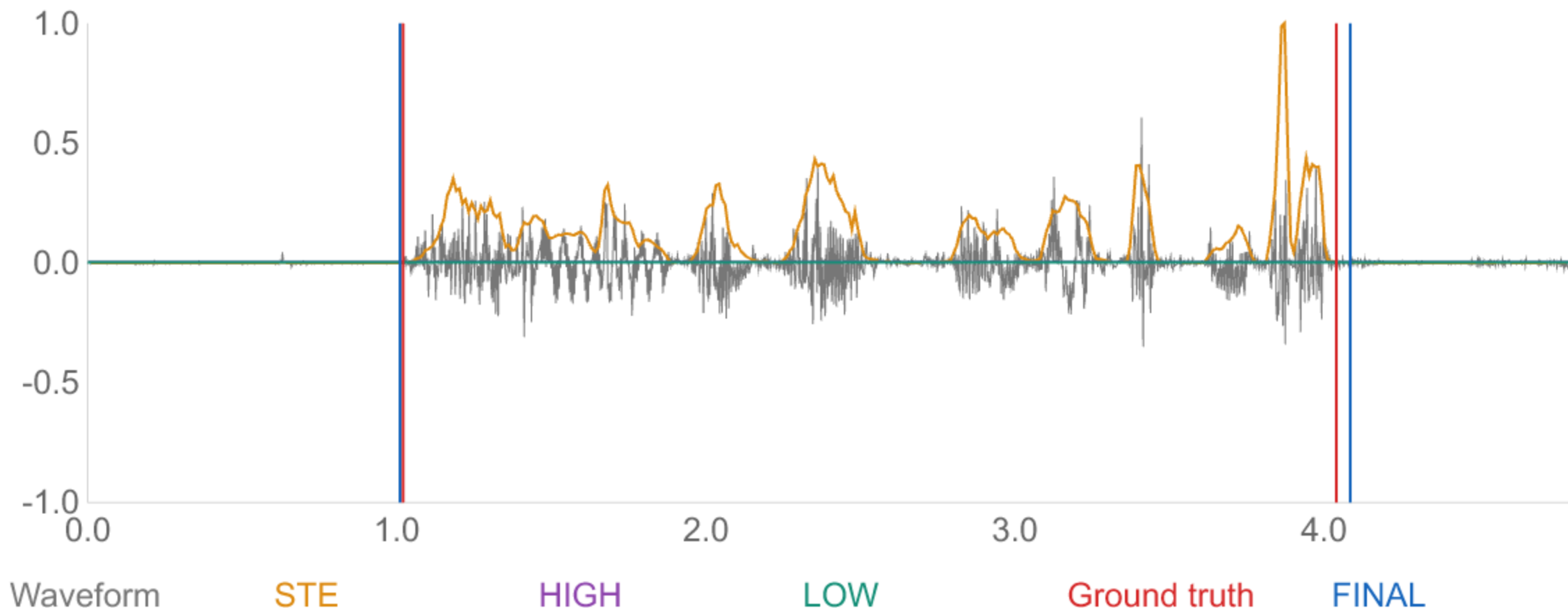


5. Hysteresis: merge <200ms gaps, discard <100ms speech

Learned threshold: 0.002877734

phone_F2: test results

MAE 27.50 ms RMSE 32.60 ms

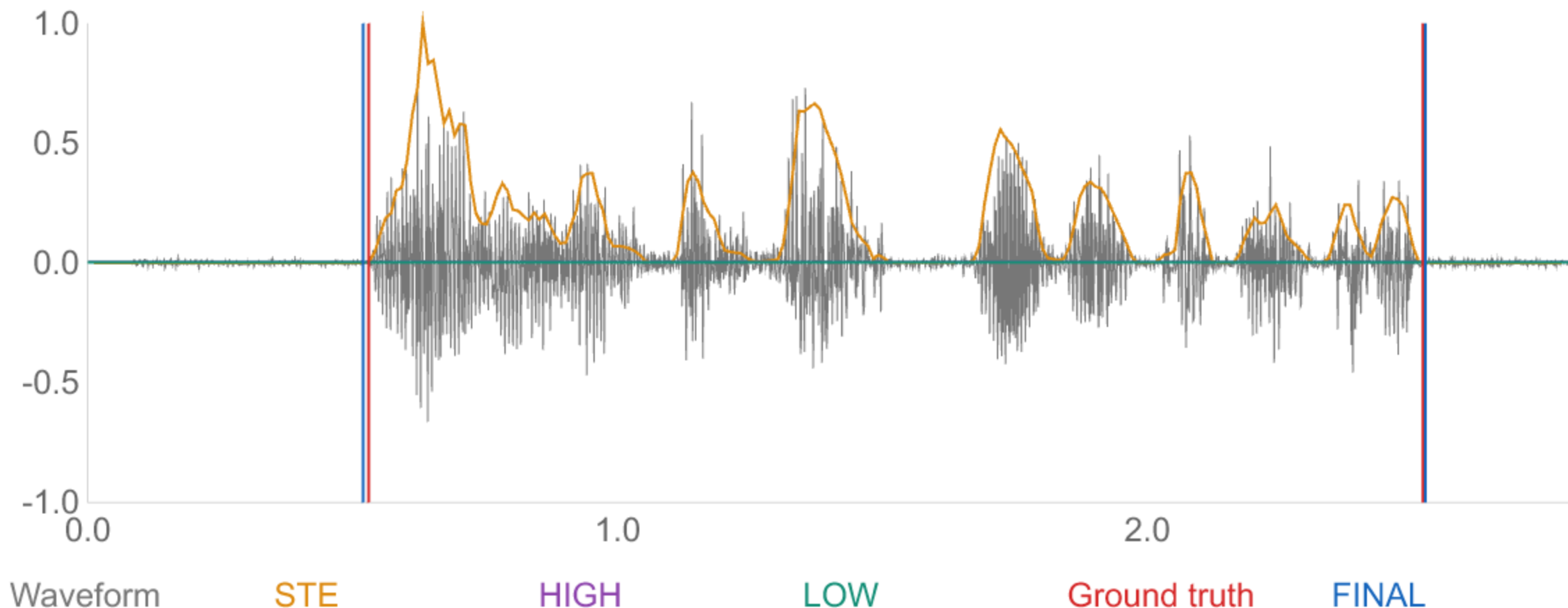


Time (s)

Noise tails can extend speech.

phone_M2: test results

MAE 7.50 ms RMSE 7.91 ms

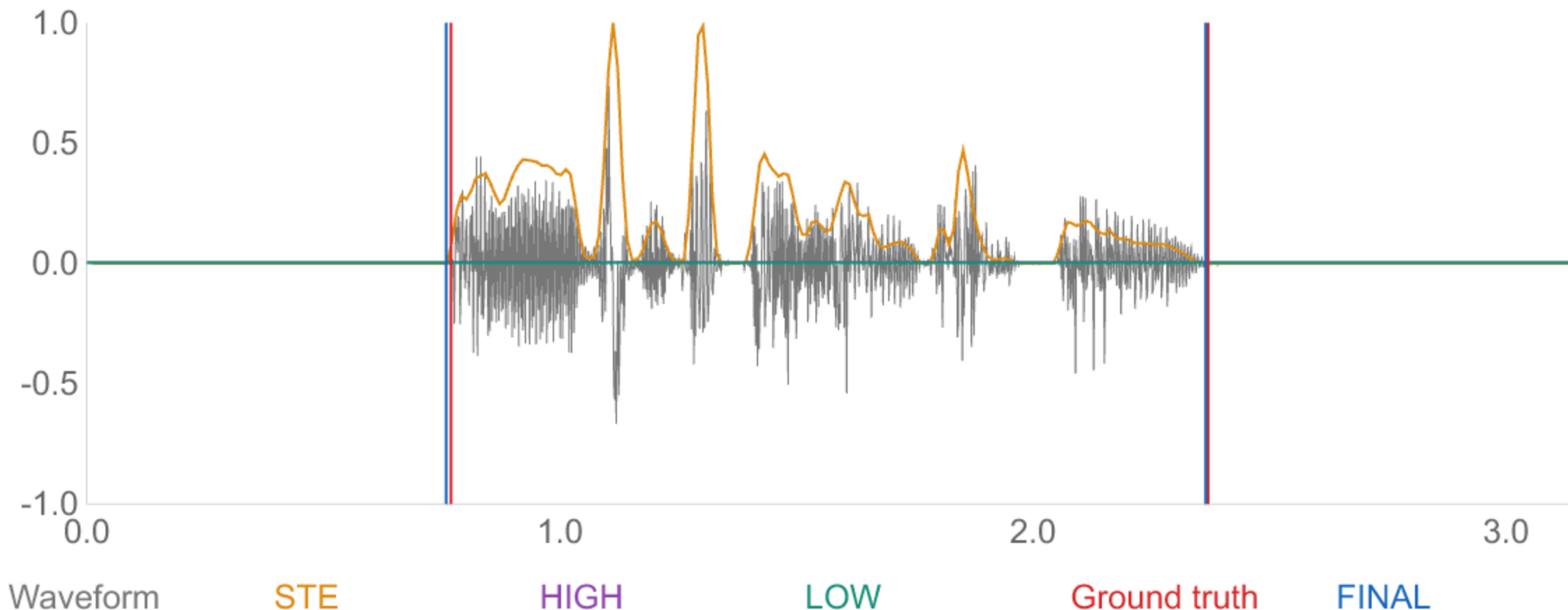


Time (s)

Noise tails can extend speech.

studio_F2: test results

MAE 7.51 ms RMSE 7.91 ms

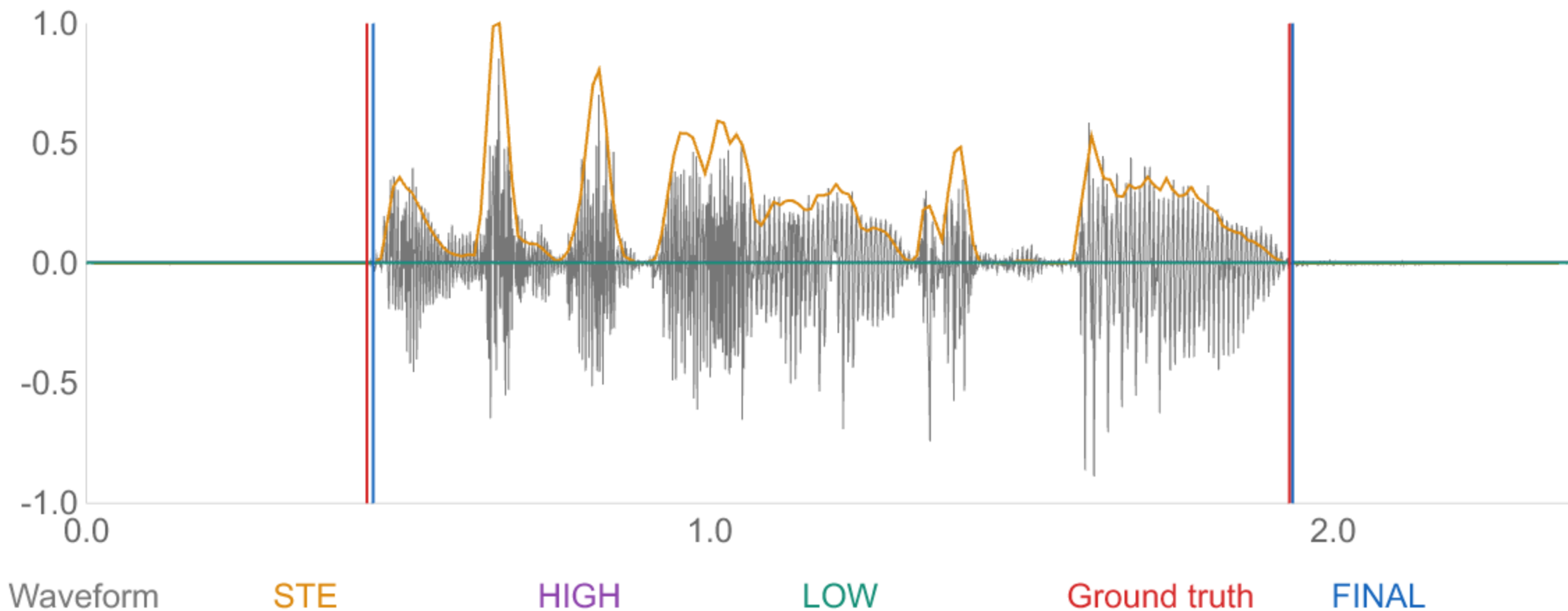


Time (s)

Framing and thresholds affect boundary error.

studio_M2: test results

MAE 7.49 ms RMSE 7.90 ms



Time (s)

Framing and thresholds affect boundary error.

Error on four test files

File	MAE (ms)	RMSE (ms)
phone_F2	27.50	32.60
phone_M2	7.50	7.91
studio_F2	7.51	7.91
studio_M2	7.49	7.90

Mean MAE: 12.50 ms

Gaussian fit depends on training data.